

Assignment 7 - Baseline Models

Empirical reference point for the engineering thesis on hospital service use among GP practice populations in Scotland

Target variable: hospital_use_per_1000 | seed: 42 | primary metric: holdout_rmse

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1. Purpose of the Baseline Stage

The purpose of Assignment 7 is to establish a controlled empirical reference point against which more advanced models in Assignment 8 can be evaluated. The research logic remains consistent with the previous assignments: the analysis tests whether the demographic structure of GP practice populations improves the explanation of variation in hospital service use compared with a simpler model based only on practice size and GP availability. The dependent variable is continuous, so the task is treated as a regression problem rather than a patient-level risk classification problem.

Assignment 2 showed that the literature on hospital admissions is methodologically heterogeneous. Some studies operate at the patient level and report AUC or C-statistics, whereas the explanatory strand at the practice or regional level more often relies on regression models and interpretation of demographic, deprivation, and accessibility factors. For this reason, the baseline stage prioritises transparent models and one classical nonlinear benchmark instead of maximising algorithmic complexity.

2. Model Selection and Configuration

Four configurations were evaluated. The dummy mean baseline predicts the training-set mean and provides a lower reference point. OLS F0 is the main baseline defined in Assignments 1 and 3: it uses only log-transformed practice size and GP availability. Ridge F1 extends F0 with the share of patients aged 65+, female share, deprivation, and HB/HSCP regional controls; it therefore represents the core hypothesis about the added value of demographics while using regularisation to reduce instability under collinearity. Decision Tree F1 is a classical machine learning benchmark that checks whether nonlinear splits and interactions provide additional benefit over interpretable regression models.

The hyperparameter strategy was deliberately limited. OLS and the dummy model were not tuned. Ridge used a small alpha grid of 0.1, 1.0, and 10.0, selected only through group-aware validation on the training partition. The decision tree used a conservative pre-specified maximum depth and minimum leaf size. This is appropriate for a baseline stage: the goal is to define a fair reference point, not to over-optimize models on a tiny sample.

Table 1. Baseline model configurations.

Model	Features	Parameters	Encoded feature count	Strategy
Dummy mean baseline	F0	{}	2	No tuning; predicts the training-set mean.
OLS baseline	F0	{"alpha": 0.0}	2	No tuning; ordinary least squares.
Ridge extended baseline	F1	{"alpha": 0.1}	10	Limited alpha grid selected by group-aware CV on training folds only.
Decision Tree benchmark	F1	{"max_depth": 3, "min_leaf": 2}	10	No tuning; conservative tree depth and leaf size fixed before test evaluation.

3. Controlled Training Protocol

Preprocessing follows the decisions formalised in Assignment 6. QF quality columns are excluded from the predictor set, PracticeCode is used only for grouping and traceability, AllAges is replaced by log1p(AllAges), and demographic structure is represented by a compact set of shares rather than many overlapping counts. Numeric missing values are imputed with the median, categorical values with the most frequent category, and HB/HSCP are one-hot encoded. All stateful transformations are fitted only inside the relevant training fold.

The final split is PracticeCode-aware: the test practice or practices (1004) do not appear in the training partition. This protects the evaluation from leaking practice-specific structure between train and test data. Assignment 3 planned 5-fold cross-validation; because the repository sample has only 4 practices, validation uses 3 valid group folds on the training partition. This is a technical adjustment to the demonstration sample, not a change in the target protocol for the full dataset.

4. Performance Results

Table 2. Holdout and group-validation results.

Model	Features	Test R2	Test RMSE	Test MAE	CV R2	CV RMSE	CV MAE
Dummy mean baseline	F0	-818.568	52.632	52.600	-1039.550	45.871	45.833
OLS baseline	F0	-16.617	7.717	7.539	-686.193	43.961	43.949
Ridge extended baseline	F1	-18.357	8.089	8.083	-38.230	8.314	8.294
Decision Tree benchmark	F1	-438.536	38.544	38.500	-2289.320	57.627	57.589

R2 measures the proportion of target variance explained by the model; negative values are possible when a model predicts worse than the test-set mean. RMSE is the square root of the mean squared prediction error and therefore penalises larger mistakes more strongly. MAE reports the average absolute error. Consistent with Assignments 1 and 3, all relevant metrics are reported together and no selective metric reporting is used.

The lowest test RMSE was achieved by OLS baseline (7.717). This corresponds to an RMSE change of 0.00% relative to OLS F0 and an R2 difference of 0.000.

5. Critical Interpretation

OLS F0 achieved an RMSE of 7.717 and an R2 of -16.617. This is the most important reference point because it corresponds to the earlier definition of the thesis baseline. Ridge F1 achieved an RMSE of 8.089 and an R2 of -18.357; its result indicates whether demographic and regional controls improve performance under the same experimental procedure. The interpretation must remain cautious because the repository sample is a synthetic pipeline check rather than the full empirical thesis dataset.

Decision Tree F1 achieved an RMSE of 38.544 and an R2 of -438.536. If the nonlinear benchmark does not consistently outperform the regression models, this is aligned with Assignment 2: in the practice-level problem, the priority is a transparent estimate of the added value of demographics, not replacing an interpretable model with a black box. If the tree improves error on the full dataset, the result should be treated as evidence that Assignment 8 should examine interactions while preserving the same split, metrics, and leakage controls.

The main lesson from the baseline stage is not merely the ranking of models, but the empirical freezing of comparison conditions. Assignment 7 establishes reference values for F0, F1, and the nonlinear benchmark. In Assignment 8, every claim of improvement should be measured against these results, preferably through R2 difference, percentage RMSE reduction, and stability under group-aware validation. On the full dataset, the success criterion should remain at least a 5 percentage-point improvement in R2 and at least a 5% reduction in RMSE relative to OLS F0.

6. Limitations

The current results are reliable as a methodological test, but not as a substantive conclusion about the Scottish health care system. The repository sample contains 12 observations and 4 practices, so R2 values can be unstable and cross-validation must be limited to the number of available groups. This decision is recorded in the metadata

and does not contaminate the test set. Once the full Public Health Scotland analytical table is available, the same implementation should be rerun without changing feature definitions, metrics, or hyperparameter-selection rules.